Pierre Mikhiel

Creative Technologist / UX Engineer - AI, Real-Time 3D, Browser-Based Interactivity

Abu Dhabi, UAE | Open to New York or Redmond relocation | +971 55 984 5520 | pmk2057@nyu.edu | pierremikhiel.com | github.com/zuett

PROFILE

Creative technologist and UX engineer building functional prototypes, browser-based AI/ML interactions, and real-time Unity/XR experiences for research, education, and public storytelling. Strong at translating ambiguous creative concepts into high-fidelity demos, validating experience quality through user research and QA, and collaborating across design, engineering, research, production, and media teams.

TECHNICAL SKILLS

Creative engineering: JavaScript, HTML/CSS, p5.js, ml5.js, TensorFlow.js, Python, C#, C++, Arduino

Real-time / immersive: Unity 3D, XR Interaction Toolkit, Cinemachine, Blender, VR prototyping, motion capture, projection, RGB DMX, QLC+

AI / ML interaction: HandPose, FaceMesh, Body Pose, Body Segmentation, dataset testing, model behavior tuning, in-browser computer vision

UX / product storytelling: high-fidelity prototyping, interaction design, demo scripting, visual pacing, QA testing, audience-facing installations, technical documentation

CREATIVE TECHNOLOGY EXPERIENCE

Machine Learning Research Assistant - ml5.js, Interactive Media NYUAD

Abu Dhabi, UAE | Jan. 2026 - Present

- Contribute to the creative application of ml5.js, an open-source browser-based machine learning library built on TensorFlow.js, by prototyping interactive demos for artists, educators, and students.
- Proposed and co-developed ASL Trainer, a deployed webcam-based fingerspelling practice tool using p5.js, ml5.js, HandPose, embedded neural assets, real-time feedback, and full-screen learning modes.
- Designed UI/UX flows for A-Z practice, word mode, reference visuals, progress feedback, and onboarding; tuned recognition behavior and in-browser performance for smoother live interaction.
- Built Interactive ML Portrait Builder, a browser tool that lets users compose Face Mesh, Body Pose, Hand Pose, Body Segmentation, and Face Texture models into exportable starter projects.

Research Assistant - VR Emotion Induction Toolkit, Aim Lab NYUAD

Abu Dhabi, UAE | May 2025 - Sep. 2025

- Led design and Unity implementation for the Anger module of a modular VR toolbox studying universal emotions and stress.
- Engineered interactive scenarios using social unfairness, time pressure, environmental pacing, animation, and spatial cues to evoke measurable emotional responses.
- Supported biometric and behavioral validation workflows, connecting immersive experience design with research-grade measurement and iteration.

Co-Author - vER: Virtual Human Companionship for Preoperative Care

IMX ACM 2025; EMRN Barcelona 2025

- Designed virtual companion agents that use social presence to reduce anxiety in simulated pre-operative hospital settings.
- Validated agent experience through a pilot study combining biometric tracking, qualitative surveys, and post-study analysis.

Creative Technology / Unity QA Intern, Mental.ae

Abu Dhabi, UAE | Nov. 2024 - Jan. 2025

- Performed QA testing on interactive food-safety training modules, documenting UX issues, navigation friction, and product behavior gaps for design revisions.
- Used Unity Cinemachine to create dynamic camera systems and cinematic sequences for immersive training simulations and demo-ready learning content.

SELECTED PROTOTYPES, DEMOS & INSTALLATIONS

- **V!Real(ity) - Capstone interactive installation + VR:** Designed and engineered a room-scale sensory system combining Unity VR, RGB DMX lighting, layered composite visuals, primed audio, and multi-user shadow mechanics to turn an abstract perception thesis into a filmable, audience-facing experience.
- **Mondrian's Mirage - VR installation, Louvre Abu Dhabi:** Translated a Mondrian composition into a navigable Unity-based VR environment with 3D spatial transitions, abstract architectural forms, and embodied museum interaction.
- **Whispers from the Depth - XR performance, NYUAD Black Box Theater:** Built an environmental XR performance about marine pollution using motion capture, real-time avatar projection, immersive sound, and visual storytelling for a live audience.
- **Beat-Hoven - Interactive music game, p5.js + Arduino:** Created a physical-digital music interface with Arduino buttons, serial communication, synchronized p5.js visuals, and layered beatboxing loops for public showcase interaction.
- **Alien Meta Rave / Are You There? - live-digital performances, New York:** Combined VR worlds, projection, DJing, machine-learning interaction, and performer silhouettes to explore virtual presence and hybrid storytelling.

EDUCATION

New York University Abu Dhabi - B.A. in Interactive Media; Minors in Computer Science and Film & New Media

Abu Dhabi, UAE | Cum Laude | Graduated May 2026 | Global semesters: New York City and Shanghai

Relevant coursework: UX Design, Data Structures, Algorithms, VR Research and Applications, Designing Virtual Worlds, Virtual Body Performance, Sound Technology.

PUBLIC WORK & RECOGNITION

- Presented virtual companion research at IMX ACM 2025 and EMRN Barcelona 2025.
- Exhibited Unity-based VR work at Louvre Abu Dhabi and hybrid performance work at La MaMa Studio, New York.
- ArtsIT 2024 organizer/media lead; documented conference programming across photography and digital media.
- Top 10 globally in Swissnex sustainability diplomacy competition for Future of Rights of Nature.
- Residential Assistant, NYU Abu Dhabi; coordinated social and educational programming for inclusive student communities.

DEMO / PORTFOLIO LINKS

- Portfolio and project demos: pierremikhiel.com
- GitHub: github.com/zuett
- ASL Trainer source: github.com/zuett/ASL
- Interactive ML Portrait Builder source: github.com/zuett/Interactive-ML-Portrait-Builder
- V!Real(ity) capstone site: thelieyoubelieved.framer.website